

# ***Release Notes for AT32F403A\_407 Firmware Library Drivers***

## **V2.0.8-2025/11/24**

1. GPIO
  - Updated the function “gpio\_bits\_toggle()” to optimize operation procedure
2. USB
  - Updated the function “usb\_ept\_open()” to change the endpoint status from NAK to DISABLE by default when configuring synchronous IN transfer endpoints, to avoid synchronous endpoint error due to an incorrect NAK status
3. TMR
  - Updated the tmr\_pwm\_input\_config(), tmr\_encoder\_mode\_config() and tmr\_output\_channel\_config(). The channel must be disabled before configuring channel mode, otherwise it fails to switch between channel modes.
4. Updated some notes

## **V2.0.7-2025/04/25**

1. Updated RTC counter value read process.
2. Updated Flash driver and optimized process.
3. Added gpio\_bits\_toggle function.

## **V2.0.6-2024/12/13**

1. Updated CAN driver.
2. Updated Flash driver.
3. Updated some notes.

## **V2.0.5-2024/08/08**

1. Updated the “i2c\_flag\_clear” and “i2c\_init” function.
2. Updated some notes.

## **V2.0.4-2024/01/05**

1. Fixed the IAR7.4 compilation warning issue of some drivers.
2. Optimized EMAC PTP driver.

## **V2.0.3-2023/10/26**

1. Added the “interrupt\_flag\_get” function to each of the IP sections.
2. Optimized system clock switching process.

## **V2.0.2-2023/08/04**

1. Added CRC polynomial support.
2. Added “flash\_spim\_dummy\_read” and “flash\_spim\_mass\_program” functions.
3. Updated “adc\_ordinary\_channel\_set” and “adc\_preempt\_channel\_set” functions.
4. Updated some notes.

## **V2.0.1-2023/02/16**

1. Updated the value of EMAC\_MAX\_PACKET\_LENGTH to 1524.

## **V2.0.0-2022/11/18**

1. Updated the “xmc\_ext\_timing\_config” function.
2. Updated the default configuration of CRM\_HICK48\_NODIV when HICK is selected as PLL clock source.